

EMCH 260

SOLID MECHANICS

Ming Hu

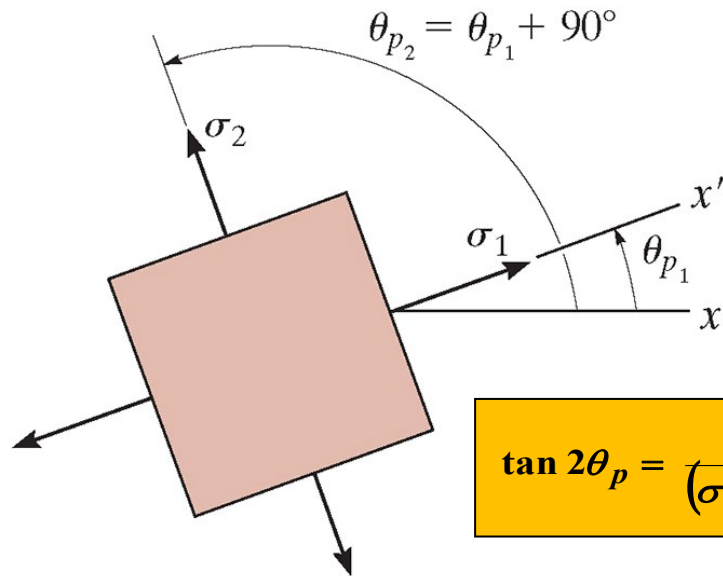
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Stress Transformation

Chapter 9.2

- ✓ Understand Mohr's circle – plane stress
- ✓ Know how to use the Mohr's circle to obtain/calculate principal stresses and maximum in-plane shear stress

PREVIOUS ON PRINCIPAL STRESSES AND MAXIMUM IN-PLANE SHEAR STRESS



In-plane principal stresses

Maximum in-plane shear stress

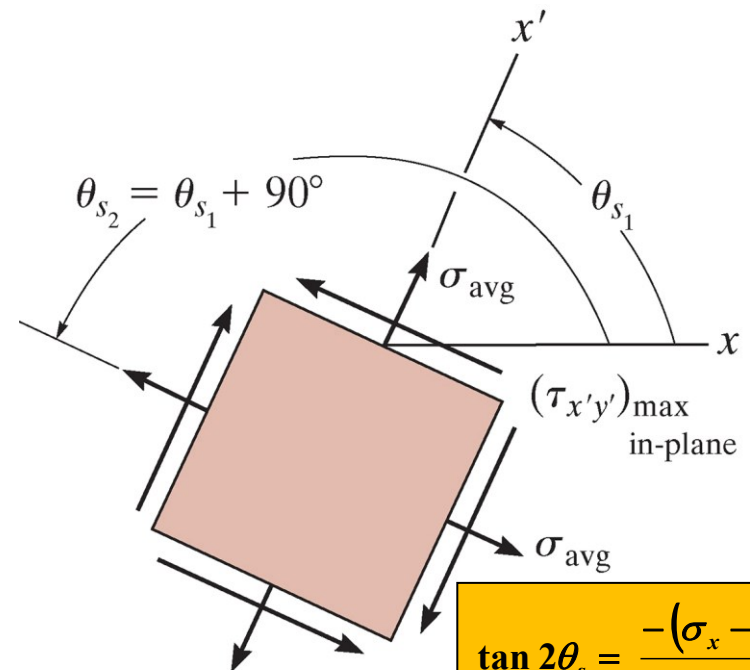
$$\tau_{in-plane}^{max} = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$\sigma_{avg} = \frac{\sigma_x + \sigma_y}{2}$$

normal stress on the planes of maximum in-plane shear stress

Principal stresses = Maximum and minimum normal stress

$$\sigma_{1,2} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$



Maximum in-plane shear stresses

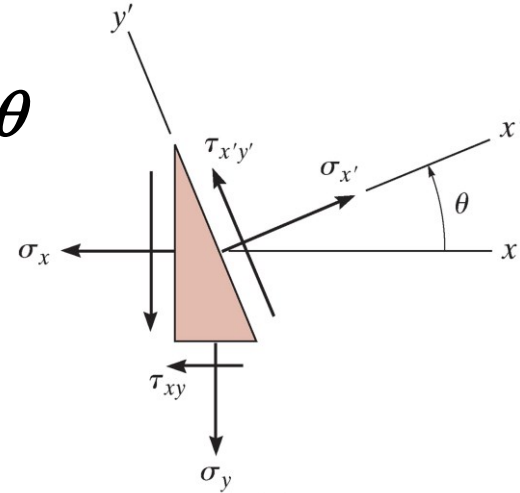
$$\tan 2\theta_s = \frac{-(\sigma_x - \sigma_y)/2}{\tau_{xy}}$$

MOHR'S CIRCLE – PLANE STRESS

We rewrite the stress component $\sigma_{x'}$ and $\tau_{x'y'}$ as follows

$$\sigma_{x'} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2} \cos 2\theta + \tau_{xy} \sin 2\theta$$

$$\tau_{x'y'} = -\frac{\sigma_x - \sigma_y}{2} \sin 2\theta + \tau_{xy} \cos 2\theta$$



Squaring each equation and adding the equation together can eliminate the parameter θ . The result is

$$\left[\sigma_{x'} - \left(\frac{\sigma_x + \sigma_y}{2} \right) \right]^2 + \tau_{x'y'}^2 = R^2 \quad R = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2} \right)^2 + \tau_{xy}^2}$$

MOHR'S CIRCLE – PLANE STRESS, CONT'D

$$\left[\sigma_{x'} - \left(\frac{\sigma_x + \sigma_y}{2} \right) \right]^2 + \tau_{x'y'}^2 = R^2$$

Since σ_x , σ_y and τ_{xy} are *known constants*, the above equation can be written in a more compact form as

$$(\sigma_{x'} - \sigma_{avg})^2 + \tau_{x'y'}^2 = R^2$$

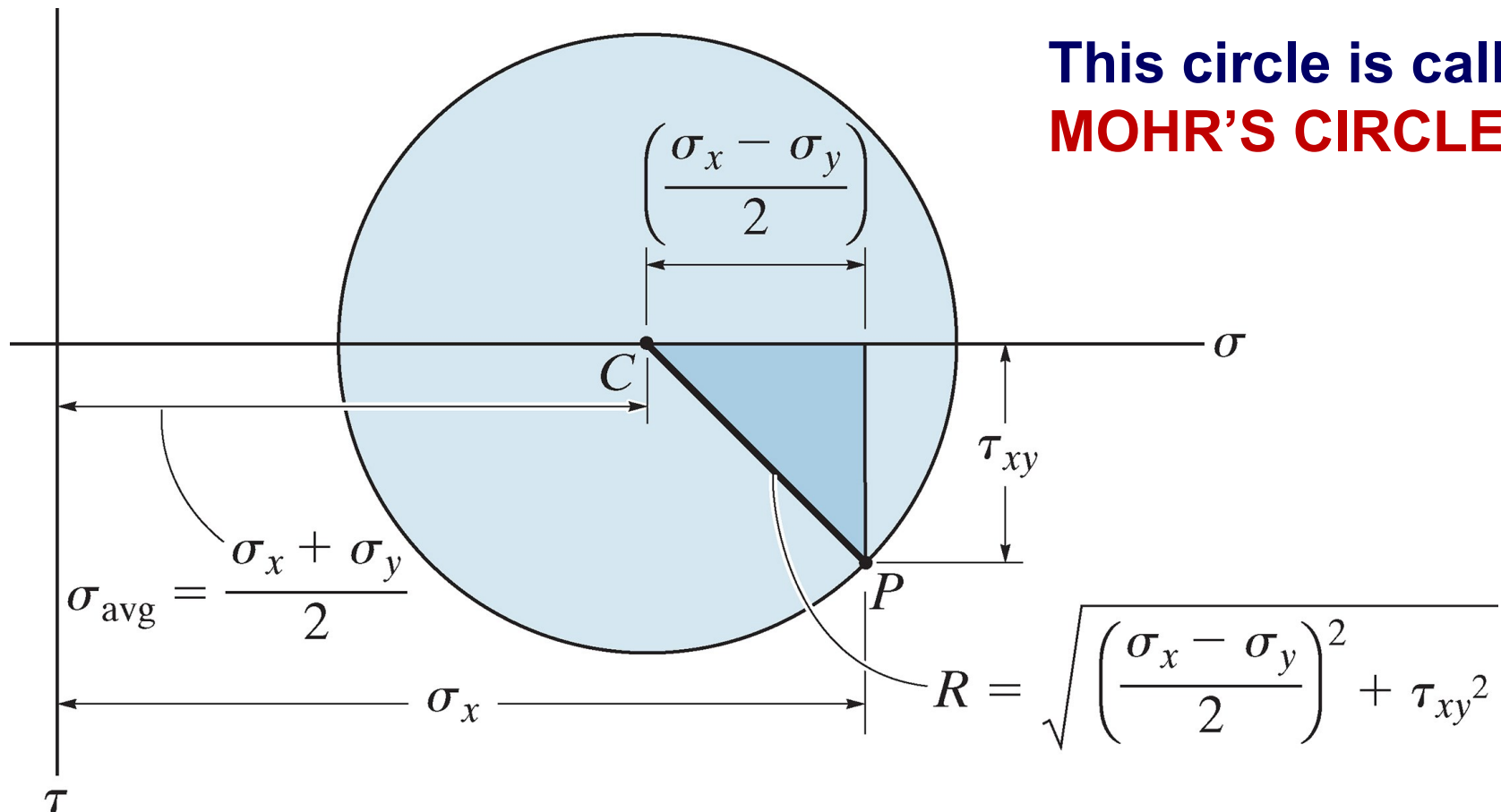
$$\sigma_{avg} = \frac{\sigma_x + \sigma_y}{2}$$

$$R = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2} \right)^2 + \tau_{xy}^2}$$

MOHR'S CIRCLE – PLANE STRESS, CONT'D

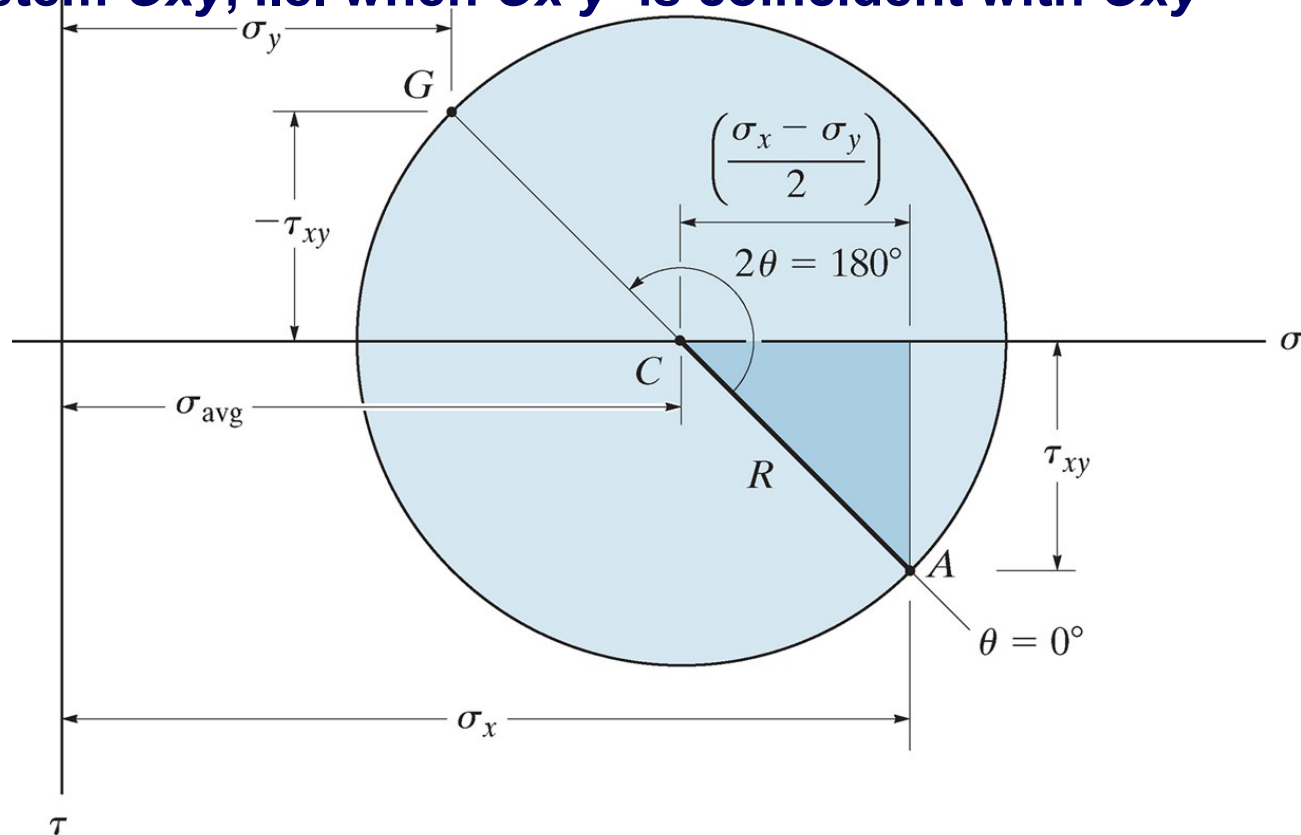
This equation represents a circle having a **radius R** and **center on σ axis at point $C(\sigma_{avg}, 0)$** as shown below

$$(\sigma_{x'} - \sigma_{avg})^2 + \tau_{x'y'}^2 = R^2$$

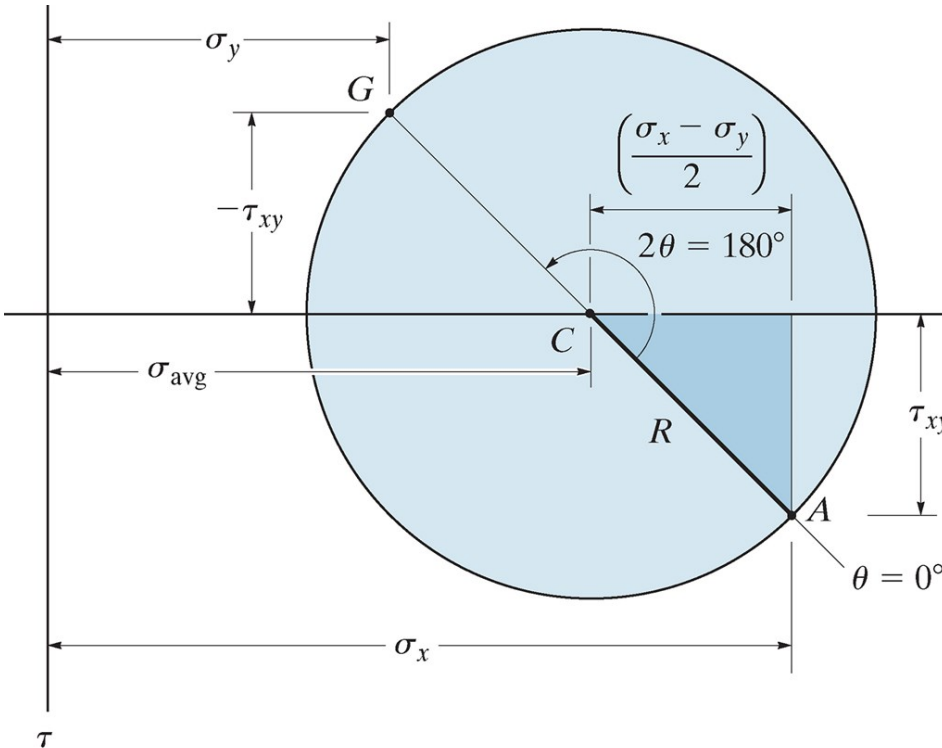


PROPERTIES OF MOHR'S CIRCLE

- Each point on Mohr's circle represents the **two** stress components $\sigma_{x'}$ and $\tau_{x'y'}$ acting on the side of the element defined by the outward x' axis (i.e. in the **new** coordinate system $Ox'y'$), when this axis is in a specific direction θ .
- Reference point A (σ_x, τ_{xy}): stress state in the **original** coordinate system Oxy , i.e. when $Ox'y'$ is coincident with Oxy

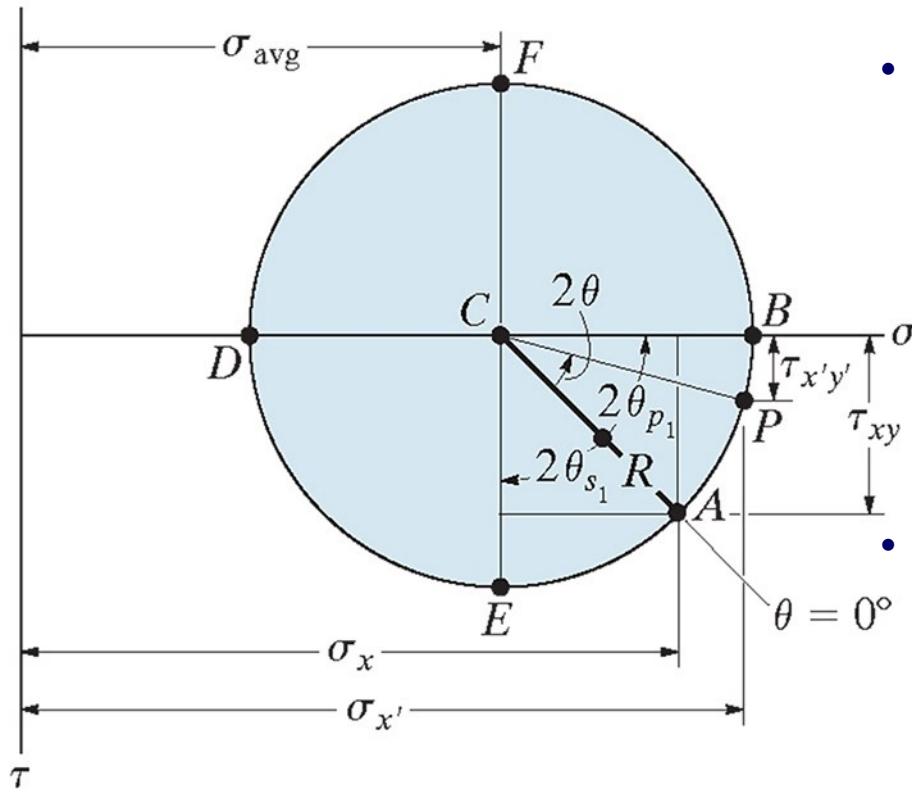


PROPERTIES OF MOHR'S CIRCLE, CONT'D



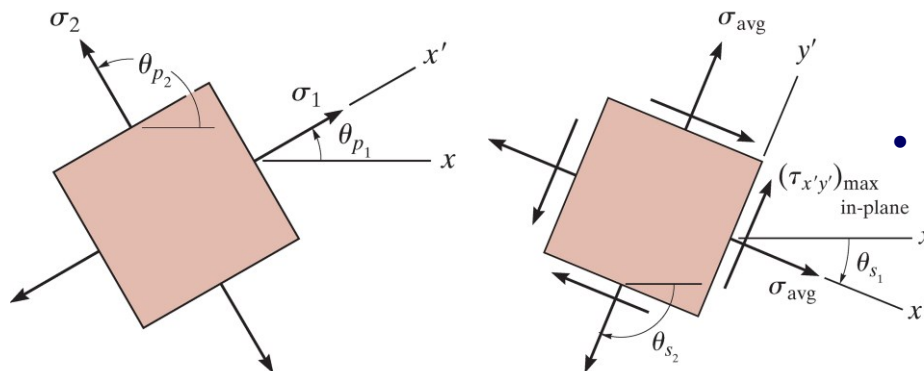
- Reference line CA : corresponding to original orientation of the coordinate system Oxy , i.e. $\theta = 0^\circ$.
- A rotation θ of the x' axis on the element corresponds to a rotation 2θ on the circle in the **same direction**
- Radial line CG (opposite to the reference line CA): the point G represents the **two** stress components σ_y and $-\tau_{xy}$ acting on the side of the element in the original coordinate system Oxy

PROPERTIES OF MOHR'S CIRCLE, CONT'D



- Points **B** and **D**: the principle stresses σ_1 and σ_2 ($\sigma_1 \geq \sigma_2$), where the Mohr's circle intersects the σ -axis ($\tau = 0$). The principle planes are determined by angles θ_{p1} (represented on the circle as $2\theta_{p1}$ in **counterclockwise** direction) and θ_{p2} .

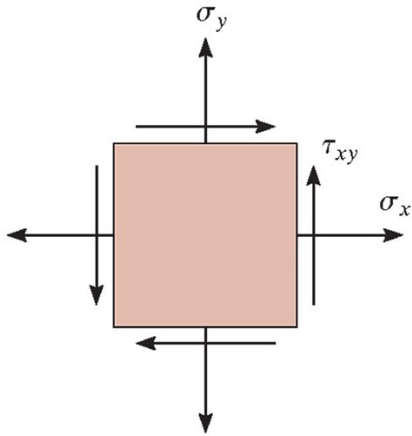
- Point **E** or **F**: the average normal stress and maximum in-plane shear stress. The orientation is determined by angles θ_{s1} (represented on the circle as $2\theta_{s1}$, **NOTE: clockwise!**) and θ_{s2}



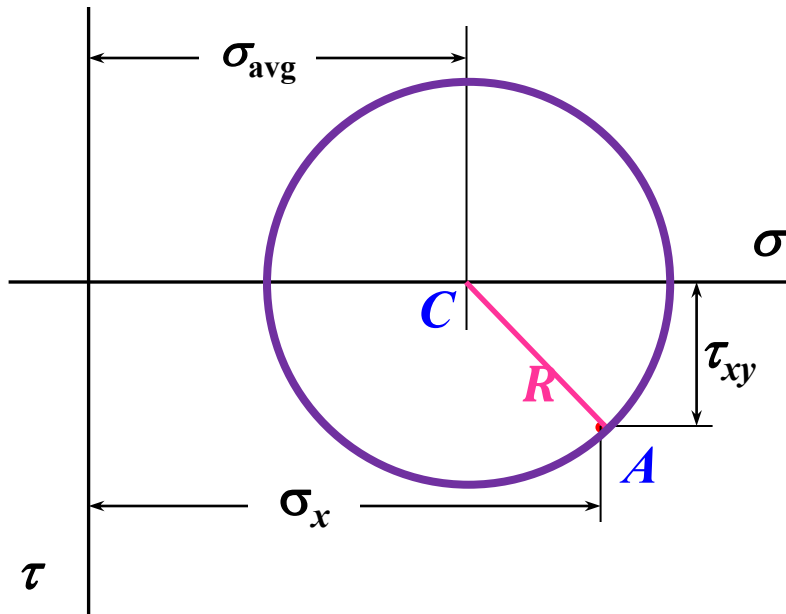
- Any point **P**: an arbitrary plane with orientation of angle θ and stress components ($\sigma_{x'}$ and $\tau_{x'y'}$)

PROCEDURE FOR ANALYSIS

Step 1: Construction of Mohr's circle



A stress state of a point for which all stress components σ_x , σ_y and τ_{xy} are positive (just for example)



- Establish a coordinate system;
 σ - τ axis
- Plot the center of the Mohr's circle
 $C(\sigma_{avg}, 0)$ on σ -axis $\sigma_{avg} = (\sigma_x + \sigma_y)/2$
- Plot the reference point $A(\sigma_x, \tau_{xy})$.
This represents $\theta = 0^\circ$
- Connect point A with the center C ,
and CA becomes the radius of the
circle
- Sketch the circle

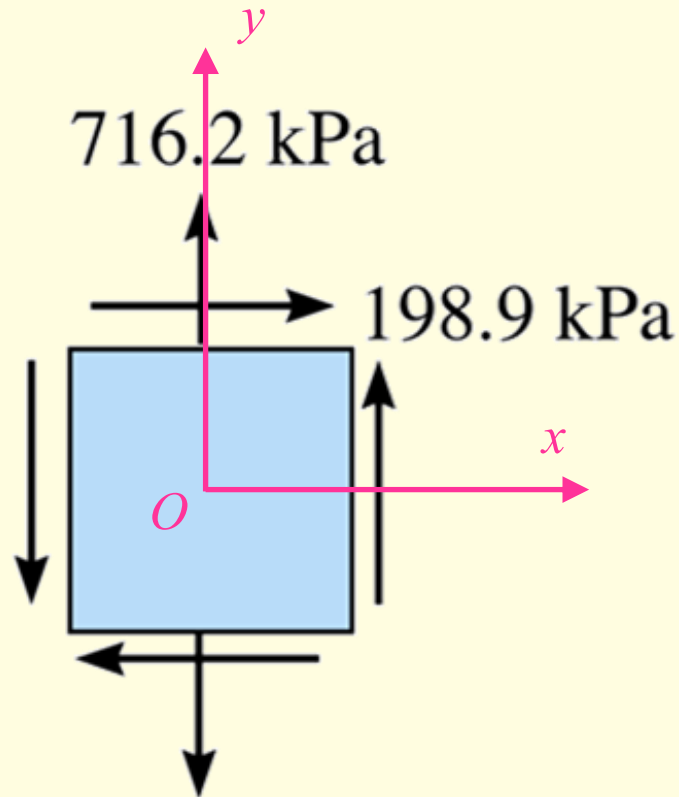
PROCEDURE FOR ANALYSIS, CONT'D

Step 2: Use the Mohr's circle

With the Mohr's circle, all properties can be known and calculated, e.g. principle stresses, maximum in-plane shear stress, and stresses on arbitrary plane.

IN-CLASS EXAMPLE

The plane-stress state of a point is represented by the element with normal stress and shear stress shown. Determine the principal stresses and the maximum in-plane shear stress at this point using the Mohr's circle.



IN-CLASS EXAMPLE, Cont'd

Solutions

Step 1: Construction of the Mohr's Circle

Given $\sigma_x = 0$, $\sigma_y = 716.2$ kPa, $\tau_{xy} = 198.9$ kPa, the center of the circle C is located on the σ -axis, at the point

$$\sigma_{\text{avg}} = \frac{\sigma_x + \sigma_y}{2} = 358.1 \text{ kPa}$$

The point $C(358.1, 0)$ and the reference point $A(0, 198.9)$ are labeled. Then, the radius of the circle is $R = \sqrt{358.1^2 + 198.9^2} = 409.6$, and the circle is constructed.

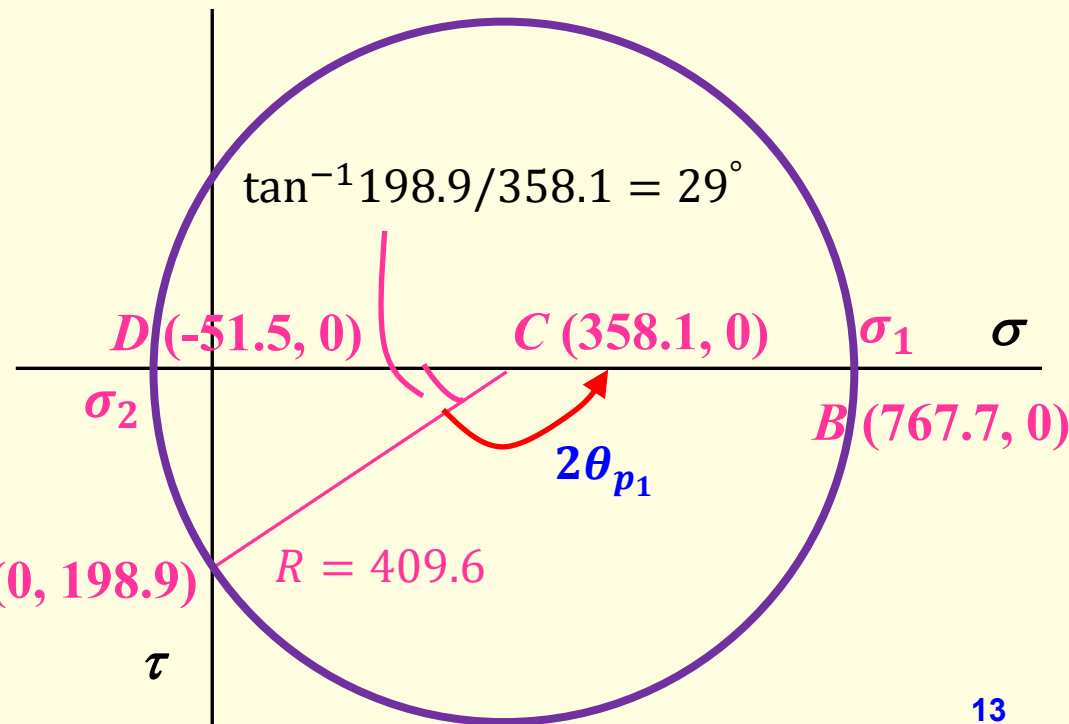
Step 2: Principal Stresses

The principal stresses are the two intersection points of the circle with the σ -axis:

$$\sigma_1 = 358.1 + R = 767.7 \text{ kPa}$$

$$\sigma_2 = 358.1 - R = -51.5 \text{ kPa}$$

The angle θ_{p_1} measured counterclockwise from CA to CB can be found from the circle as
 $2\theta_{p_1} = 180^\circ - 29^\circ$, i.e. $\theta_{p_1} = 75.5^\circ$



IN-CLASS EXAMPLE, Cont'd

Solutions, Cont'd

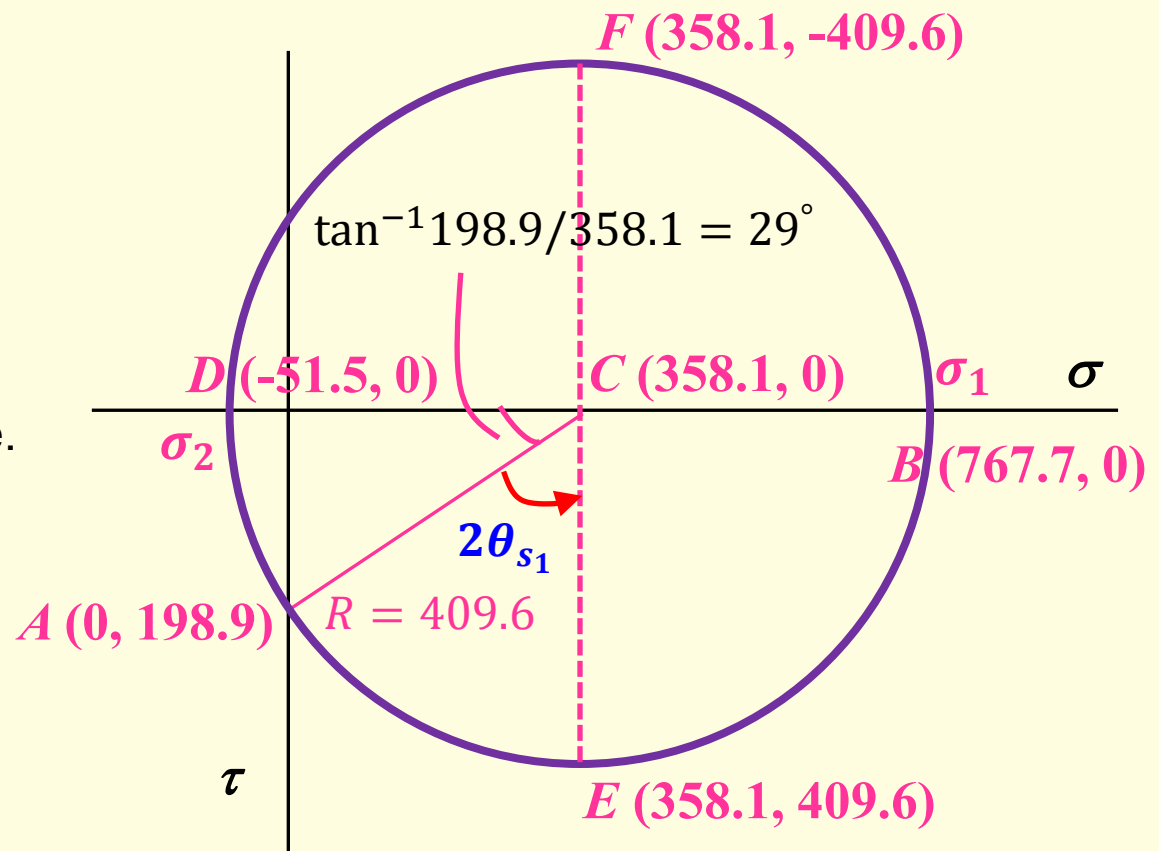
Step 3: Maximum In-Plane Shear Stress

The maximum in-plane shear stress and the average normal stress are identified by point *E* or *F* on the circle:

$$\sigma_{\text{avg}} = \frac{\sigma_x + \sigma_y}{2} = 358.1 \text{ kPa}$$

$$\tau_{\text{in-plane}}^{\text{max}} = R = 409.6 \text{ kPa}$$

The angle θ_{s_1} measured counterclockwise from *CA* to *CE* can be found from the circle as $2\theta_{s_1} = 90^\circ - 29^\circ$, i.e.
 $\theta_{s_1} = 30.5^\circ$



RELEVANT PRACTICE

- **Assignment 3 on Blackboard, problem #3 (go to Blackboard under the “Assignments” folder, **Assignment 3 due: 8:30 AM, Friday, Nov. 5th 2021**)**
- **Optional Extra Exercise 3 on Blackboard, problem #3 (go to Blackboard under the “Extra Exercises” folder, **Extra Exercise 3 due: 8:30 AM, Friday, Nov. 5th 2021**)**

OUT-OF-CLASS READING

- **Textbook, pp. 469 – 475**
 - **In particular, Examples 9.7 – 9.9**